Paper 10 - Working English translation

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HODGE THEORY II

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Introduction

[Polished translation of the opening paragraphs]

This paper develops the formalism of filtrations and then uses it to construct mixed Hodge structures on the cohomology of smooth algebraic varieties that need not be compact. The key point is that, although the cohomology of a nonsingular quasiprojective variety is no longer pure, it can still be organized as a successive extension of pure Hodge structures whose weights lie between n and 2n.

The proof is essentially algebraic. It combines classical Hodge theory with Hironaka-style resolution of singularities, which allows one to express the cohomology of a smooth quasiprojective variety, via a spectral sequence, in terms of the cohomology of smooth projective varieties.

Section 1 collects some general facts on filtrations and proves two key results: the theorem on mixed Hodge structures, used later through its corollary on strictness, and the lemma of the two filtrations. Section 2 recalls classical Hodge theory and introduces mixed Hodge structures. The heart of the paper is Section 3.2, where the mixed Hodge structure on H^*(X,C) is defined and the fundamental spectral-sequence degenerations are proved. Section 4 gives applications, especially the fixed-part theorem, semisimplicity, and consequences for homomorphisms of abelian schemes.

[Source-aligned literal body continues]

After Hodge, l'espace of cohomologie H n (X, C) d'une variety kahlerienne compact X is muni d'une << Hodge structure >> of weights n, i.e. d'une bigraduation natural

$$H^n(X, C) = \bigoplus_{p+q=n} H^{pq}$$

verifiant H pq = H qp . On shows ici que the cohomology complexe d'une algebraic nonsingular variety, non necessairement compact, is munie d'une structure d'espece a peu more generale, qui fait apparaitre H n (X, C) as << extension successive >> of Hodge structures of weights decroissants, contenus entre 2n and n, dont the nombres of Hodge hpq = dim H pq are zero for p > n or q > n.

Le lecteur trouvera expose in [14] the yoga qui sous-tend cette construction.

Travail presente as these of doctorat a l'Universite d'Orsay.

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La proof, essentiellement algebrique, repose d'une part on the Hodge theory, d'autre part on the resolution of singularities a the Hironaka qui permet, via a spectral sequence, << d'exprimer >> the cohomology d'une algebraic nonsingular variety quasi-projective en terme of the cohomology of nonsingular projective varieties.

Le SS 1, outre quelques sorites on the filtrations reunis for the commodite of the lecteur, contient deux resultats-clefs :

a) Le theorem (1.2.10), qui ne will be utilise que via son corollary (2.3.5), donnant the properties fondamentales of << Hodge structures mixtes >>.

b) Le << lemma of deux filtrations >> (1.3.16).

Le SS 2 recalls the Hodge theory and introduces the Hodge structures mixtes. Le coeur of ce travail is the no 3.2, qui defines the Hodge structure mixte of H ∗ (X, C) and etablit quelques degenerations of spectral sequences.

Le SS 4 gives diverses maps, all deduites of (4.1.1) and of the theorie of the (K/A)-trace for the Hodge structures qui en resulte (4.1.2). Les principales are (4.2.6) and (4.4.15).

1. Filtrations

1.1. Filtered objects. (1.1.1) Let A a categorie abelienne.

On aura a considerer of filtrations of type Z, en general finies, on the objects of A :

Definition 1.1. Une filtration decroissante (resp. croissante) F d'un object A of A is a famille (F n (A))n∈Z (resp. (Fn (A))n∈Z) of sous-objets of A, verifiant

n, m n m ⇒ F n (A) ⊃ F m (A)

(resp. n, m n m ⇒ Fn (A) ⊂ Fm (A)).

Un object filtre is a object muni d'une filtration. Quand il n'y aura pas danger of confusion, one designera often par a meme lettre of filtrations on of objects differents of A .

If F is a filtration decroissante (resp. croissante) on A, one pose F (A) = 0 and F -∞ (A) = A (resp. F-∞ (A) = 0 and F∞ (A) = A).
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Les filtrations decalees d'une filtration decroissante W are definies par

W[]n = W n-[]   (n, ` ∈ Z).

(1.1.3) If F is a filtration decroissante (resp. croissante) of A, then the Fn (A) = F -n (A) (resp. the F n (A) = F-n (A)) forment a filtration croissante (resp. decroissante) of A. Ceci permet en principe of ne considerer que of filtrations decroissantes ; sauf mention explicite of the contraire, par << filtration >> one entendra dorenavant << filtration decroissante >>.

(1.1.4) Une filtration F of A is dite finie s'il existe n and m tels que F n (A) = A and F m (A) = 0.

(1.1.5) Un morphism d'un object filtre (A, F) in a object filtre (B, F) is a morphism f of A in B qui satisfies f (F n (A)) ⊂ F n (B) for n ∈ Z.

Les objects filtres (resp. filtres of filtration finie) of A forment a categorie additive in laquelle existent the limites inductives and projectives finies (and therefore the noyaux, conoyaux, images and coimages d'un morphism).

Un morphism f : (A, F) → (B, F) is dit strict, or strictement compatible aux filtrations, if the fleche canonical of Coim(f) in Im(f) is a isomorphism d'objets filtres (cf. (1.1.11)).

(1.1.6) Let 0 the functor contravariant identique of A in the categorie duale A 0 . If (A, F) is a object filtre of A , the (A/F n (A))0 s'identifient a of sous-objets of A0 . La filtration on A0 duale of F is definie par

F n (A0) = (A/F -n (A))0 .

Le bidual of (A, F) s'identifie a (A, F). Cette construction identifie the duale of the categorie of objects filtres of A a the categorie of objects filtres of A 0 .

(1.1.7) If (A, F) is a object filtre of A , son gradue associe is l'objet of A Z defini par

Grn (A) = F n (A)/F n+1 (A).

La convention (1.1.6) is justifiee par the formule simple

Grn (A0) = Gr-n (A)0

qui se satisfies on the diagramme autodual

A/F -n (A) ←- A/F -n+1 (A)
| |
(1.1.7.1)   Grn (A)   F n (A)
| . |
F n+1 (A) -→   0

(1.1.8) Let (A, F) a object filtre and j : X ,→ A a sous-objet of A. La filtration induite on X par F is l'unique filtration on X telle que j let strictly compatible aux filtrations ; one a

F n (X) = j -1 (F n (A)) = X ∩ F n (A).

Dualement, the filtration quotient on A/X (unique filtration telle que p : A → A/X let strictly compatible aux filtrations) is given par

F n (A/X) = p(F n (A)) = (X + F n (A))/X = F n (A)/(X ∩ F n (A)).

Lemme 1.2. If X and Y are deux sous-objets of A, with X ⊂ Y , then on Y /X = Ker(A/X → A/Y), the filtration quotient of celle of Y coincide with celle induite par celle of A/X. Dans the diagramme

A/Y   ←-   A/X
| |
Y /X

|

X -→ Y -→ A

the fleches are strictes.

(1.1.10) On appellera the filtration (1.1.9) on Y /X the filtration induite par celle of A. After (1.1.9), sa definition is autoduale.

En particulier, if $S : A \xrightarrow{f} B \xrightarrow{g} C$ is a 0-sequence, and if B is filtre, then H(S) = Ker(g)/=(f) = Ker(Coim(g) → Coim(g)) is muni d'une filtration induite canonical.

Le lecteur verifiera que :

Proposition 1.3. (i) Let f : (A, F) → (B, F) a morphism d'objets filtres of filtration finie. For que f let strict, il faut and il suffices que the sequence

0 → Gr(Ker(f)) → Gr(A) → Gr(B) → Gr(Coker(f)) → 0

let exacte.

(ii) Soit

$S : (A, F) \xrightarrow{f} (B, F) \xrightarrow{g} (C, F)$

a 0-sequence of morphisms stricts. On a then

H(Gr(S)) = Gr(H(S)).

En particulier, if S is exacte in A , then Gr(S) is exacte in A Z .

Dans a categorie of modules, dire qu'un morphism f : (A, F) → (B, F) let strict signifie que all b ∈ B, of filtration n (b ∈ F n (B)) qui is in l'image of A, is already in l'image of F n (A) :

f (F n (A)) = f (A) ∩ F n (B).

(1.1.12) If ⊗ : A1 × * * * × An → B is a functor multiadditif exact a droite, and if Ai is a object of filtration finie of Ai (1 i n), one defines a filtration on ⊗Ai , par

F (⊗Ai) = =(⊗F ki (Ai) → ⊗Ai)

ki =

(sum of sous-objets).

Dualement, if H is exact a gauche, one pose

∩

F (H(Ai)) = Ker(H(Ai) → H(Ai /F ki (Ai)))

(ki = `).

For H exact, the deux definitions are equivalent.

On etend ces definitions a of functors contravariants en certaines variables par (1.1.6). En particulier, for the functor exact a gauche Hom, one pose

F n (Hom(A, B)) = {f : A → B | `, f (F (A)) ⊂ F +n (B)}.

On a therefore

Hom((A, F), (B, F)) = F 0 (Hom(A, B)).

Sous the hypotheses precedentes, one dispose of morphisms evidents

⊗ Gr(Ai) → Gr(⊗Ai)

and

Gr(H(Ai)) → H(Gr(Ai)).

For H exact, ce are of isomorphisms inverses l'un of l'autre.

Ces constructions are compatibles a the composition of functors, en a sens qu'on laisse au lecteur the soin of ne pas expliciter.

1.2. Opposed filtrations. (1.2.1) Let A a object of A muni of deux filtrations F and F . Par definition, GrnF (A) is a quotient d'un sous-objet of A, and, en tant que tel, se trouve canonically plonge in A. On pose

$$\mathrm{Gr}^n_F \mathrm{Gr}^m_{\bar F}(A) = \mathrm{Gr}^n_F(\mathrm{Gr}^m_{\bar F}(A)).$$

C'est a sous-quotient of A, quotient of F n F (A) and sous-objet of GrnF (A).

On a of formules duales :

$$\mathrm{Gr}^n_F \mathrm{Gr}^m_{\bar F}(A) = \frac{F^n(A) \cap \bar F^m(A)}{F^{n+1}(A) \cap \bar F^m(A) + F^n(A) \cap \bar F^{m+1}(A)}.$$

(1.2.2) On appelle filtration n-opposee a a filtration F a filtration F telle que

$$Gr^p_F Gr^q_F (A) = 0 \text{ for } p + q \neq n.$$

If F admet a filtration n-opposee, one dit que F is n-opposable.

Proposition 1.4. Let F and F deux filtrations n-opposees of A. Then :

(i) F and F are finies.

(ii) For all p, $F^p(A) \oplus F^{n-p+1}(A) \xrightarrow{\sim} A$.

(iii) For all p and q with p + q = n + 1,

$$F^p(A) \oplus F^q(A) \xrightarrow{\sim} A.$$

(iv) Les gradues $Gr^p_F Gr^q_F (A)$ (p + q = n) are the facteurs d'une decomposition directe of A.

Demonstration. (i) resulte of (iii) with q = n - p + 1.

(ii) and (iii) equivalent a dire que l'application canonical

$$F^p(A) \oplus F^q(A) \to A$$

(for p + q = n + 1) is injective and surjective. L'assertion d'injectivite is equivalent a dire que $F^p(A) \cap F^q(A) = 0$, i.e. que $F^p F^q(A) = 0$. C'est a case particulier of l'hypothese. L'assertion of surjectivite is equivalent a dire que

$$F^p(A) + F^q(A) = A.$$

For the prouver, one procede par recurrence descendante on p (supposons F and F decroissantes). For p assez grand, $F^p(A) = A$ and l'assertion is claire. For passer of p a p - 1, one utilise l'exactitude of

$$0 \to Gr^{p-1}_F(A) \to A/F^p(A) \to A/F^{p-1}(A) \to 0$$

and l'isomorphisme

$$Gr^{p-1}_F(A) = F^{p-1}(A)/F^p(A) = F^{p-1} F^{n-p+1}(A).$$

L'hypothese permet d'identifier ce dernier a

$$\frac{F^{p-1}(A) \cap F^{n-p+1}(A)}{F^p(A) \cap F^{n-p+1}(A)}$$

and l'application $A/F^p(A) \to A/F^{p-1}(A)$ a l'application quotient of A par $F^{p-1}(A) + F^p(A)$. L'assertion resulte then of l'hypothese of recurrence and of l'exactitude of the sequence

$$0 \to \frac{F^{p-1}(A) \cap F^{n-p+1}(A)}{F^p(A) \cap F^{n-p+1}(A)} \to \frac{A}{F^p(A)} \to \frac{A}{F^{p-1}(A)} \to 0.$$

(iv) After (ii), the $F^p F^{n-p}(A)$ ($p \in Z$) forment a filtration of A. On a $F^p F^{n-p}(A)/F^{p+1} F^{n-p-1}(A) = Gr^p_F Gr^{n-p}_F(A)$. #

(1.2.4) Let A^{pq} a famille d'objets of A , zero sauf for a nombre fini of couples (p, q) tels que p + q = n. If one pose

$$A = \bigoplus_{p+q=n} A^{pq},$$

then, one defines deux filtrations finies n-opposees of $A = \bigoplus A^{pq}$ en posant

(1.2.1) $$F^p(A) = \bigoplus_{p' \geq p} A^{p'q'},$$

(1.2.2) $$F^q(A) = \bigoplus_{q' \geq q} A^{p'q'}.$$

On a

$$Gr^p_F Gr^q_F (A) = A^{pq}.$$

Reciproquement :

Proposition 1.5. (i) Let F and F deux filtrations finies on A. For que F and F let n-opposees, il faut and il suffices que

$$\forall p, q \quad p + q = n + 1 \Rightarrow F^p(A) \oplus F^q(A) \xrightarrow{\sim} A.$$

(ii) If F and F are n-opposees, and if one pose

$$A^{pq} = \begin{cases} 0 & \text{for } p + q \neq n \\ F^p(A) \cap F^q(A) & \text{for } p + q = n, \end{cases}$$

then $A = \bigoplus A^{pq}$, and F and F are the filtrations (1.2.4) definies par cette decomposition.

Demonstration. Prouvons (i). La condition is necessary after (1.2.3) (iii). Reciproquement, supposons-la verifiee and prove que $Gr^p_F Gr^q_F(A) = 0$ for $p + q \neq n$. Par dualite, il suffices of traiter the case or $p + q > n$. Posons $r = p$, $s = n - p + 1$ and $t = q - (n - p + 1) = p + q - n > 0$. On a $r + s = n + 1$, d'ou

$$F^r(A) \oplus F^s(A) \xrightarrow{\sim} A.$$

On en deduit que

$$F^r(A) \cap F^{s+t}(A) = F^r(A) \cap F^s(A) \cap F^{s+t}(A) = 0,$$

d'ou $Gr^p_F Gr^q_F(A) = 0$ puisque $F^p(A) \cap F^q(A) \subset F^r(A) \cap F^{s+t}(A)$.

Prouvons (ii). After (i), applique a F and F decalees, one a

$$F^p(A) \cap F^{n-p+1}(A) = 0 \quad \text{and} \quad F^p(A) + F^{n-p+1}(A) = A,$$

d'ou $A = F^p(A) \oplus F^{n-p+1}(A)$. On satisfies then que $F^p(A) = \bigoplus_{p' \geq p} A^{p',n-p'}$ and $F^q(A) = \bigoplus_{q' \geq q} A^{n-q',q'}$. #

(1.2.6) If F and F are deux filtrations n-opposees on A, one pose

$$A^{pq} = Gr^p_F Gr^q_F(A) \quad (\text{zero for } p + q \neq n).$$

After (1.2.5), one a

$$A = \bigoplus_{p+q=n} A^{pq}.$$

On appelle cette decomposition the decomposition of A en type (p, q). If $a \in A$ is of type (p, q), i.e. $a \in A^{pq}$, then

$$a \in F^p(A) \cap F^q(A).$$

Definition 1.6. Trois filtrations finies W , F and F on a object A of A are dites opposees if :

(i) For all n, the filtrations induites par F and F on $Gr^W_n(A)$ are n-opposees.

(ii) For all p, q, n, one a

$$Gr^W_n Gr^p_F Gr^q_F(A) = 0 \quad \text{for } p + q \neq n.$$

(1.2.8) Let W , F , F trois filtrations opposees on A. Posons

$$A^{pq}_n = Gr^W_n Gr^p_F Gr^q_F(A).$$

After (1.2.7) (ii), $A^{pq}_n = 0$ for $p + q \neq n$. On pose

$$A^{pq} = A^{pq}_{p+q} = Gr^p_F Gr^q_F Gr^W_{p+q}(A).$$

On a then

$$Gr^W_n(A) = \bigoplus_{p+q=n} A^{pq}$$

and the filtrations induites par F and F on $Gr^W_n(A)$ are given par (1.2.4).

(1.2.9) Let W , F , F trois filtrations opposees on A, and let W ' a quatrieme filtration. On dit que W ' is compatible au systeme (W, F, F) if elle is compatible a chacune of trois filtrations W , F , F . If W ' is compatible, then the filtrations induites par F and F on $Gr^{W'}_m(A)$ restent m-opposees, and the trois filtrations W , F , F induites on $Gr^{W'}_m(A)$ restent opposees.

Theoreme 1.7. Let A a categorie abelienne, and designons provisoirement par A_f the categorie of objects of A munis of trois filtrations finies opposees.

(i) A_f is a categorie abelienne.

(ii) Les functors << oubli of filtrations >>, Gr^W_n, Gr^p_F, Gr^q_F, $Gr^W_n Gr^p_F$, $Gr^W_n Gr^q_F$, $Gr^p_F Gr^q_F$ and $Gr^W_n Gr^p_F Gr^q_F$ of A_f in A (resp. in A^Z) are exacts.

(iii) Tout morphism $f : A \to B$ in A_f admettant a kernel (resp. cokernel, image, coimage) in A admet a kernel (resp. conoyau,

image, coimage) in A f . Le gradue of ce kernel (resp. ...) is the
kernel (resp. ...) of gradues.

(iv) Toute sequence A → B → C in A f , exacte in A , is exacte in A f.

(v) Le functor A 7→ (A, W, F, F) of A f in the categorie of objects of A munis of deux filtrations opposees admet a adjoint a gauche and a droite.

Esquisse of proof. Il suffices of prouver (iii) for the noyaux. Let f : A → B a morphism in A f . Let K a kernel of f in A . Muni of filtrations induites par W , F and F , il s'agit of voir que ces trois filtrations are opposees. La condition (1.2.7) (ii) is clearly verifiee (the Apqn are the noyaux of f : Apqn → B pqn). Verifions (1.2.7) (i). Les filtrations induites par F and F on GrnW (K) are the filtrations induites par W , F and F of K are therefore opposees and K is a kernel of f in A f . #

(1.2.11) Le theorem (1.2.10) permet of define the notion of systeme of trois filtrations opposees on a object d'une categorie abelienne A on the-quelle one ne fait no hypothese. Let en effet A ' the categorie of objects of A munis d'une filtration finie. Le functor A 7→ (A, W, F) of A f in the categorie of objects of A munis of deux filtrations finies admet a adjoint a gauche L and a droite R ((1.2.10) (v)). Le functor A 7→ (LA, W, F, F) of A f in A f is adjoint a gauche au functor identique, therefore is pleinement fidele. On en deduit que A f s'identifie a the categorie of objects of A ' munis d'une filtration W opposable.

(1.2.12) Let (A, W, F, F) a object of A f . On pose

Apqn = GrnW GrpF GrFq (A).

After (1.2.7) (ii), Apqn = 0 for p + q 6= n. On pose

Apq = Apq,p+q .

On a

$$\mathrm{GrnW}\ (A) = \bigoplus_{p+q=n} \mathrm{Apq}\ , \quad \mathrm{GrpF}\ (A) = \bigoplus_{q,n} \mathrm{Apqn}\ , \quad \mathrm{GrFq}\ (A) = \bigoplus_{p,n} \mathrm{Apqn}\ .$$

(1.2.13) Let W , F and F trois filtrations finies on A. Les conditions suivantes are equivalent :

(i) Les filtrations W , F and F are opposees.

(ii) For all n, the filtrations induites par F and F on GrnW (A) are n-opposees, and for all p, the filtrations induites par W and F on GrpF (A) are p-opposees.

(iii) For all n, the filtrations induites par F and F on GrnW (A) are n-opposees, and for all q, the filtrations induites par W and F on GrqF (A) are q-opposees.

En effet, (ii) signifie que GrnW GrpF GrqF (A) = 0 for p + q 6= n and que GrpF GrnW GrFq (A) = 0 for n + q 6= p.

De meme, (iii) signifie que GrnW GrpF GrqF (A) = 0 for p + q 6= n and que GrFq GrnW GrpF (A) = 0 for n + p 6= q.

On se ramene a supposer que S is connexe non vide, of point generique e, and que Xe is a variety abelienne simple on k(e). If (i) or (ii) is verifiee, D = End(Xe) ⊗ Q is then a corps, and, for s ∈ S, (R1 f∗ Q)s is a representation irreductible of π1 (S, s).

For prouver que (ii) ⇒ (i), one peut supposer Ye simple, and (R1 g∗ Q)s iso-typique of type (R1 f∗ Q)s . On a then a isomorphism of familles continues of Q-Hodge structures (D being of type (0, 0))

R1 g∗ Q R1 f∗ Q ⊗D Hom(R1 f∗ Q, R1 g∗ Q).

On a D ⊗ C ≃ Mn (Z ⊗ C). If e ∈ D ⊗ C se transforme par a tel isomorphism en lidempotent e11 , one a again a isomorphism of vectoriels bigradues

(R1 g∗ C)s (R1 f∗ C)s ⊗Z⊗C e(Hom(R1 f∗ C, R1 g∗ C)).

If the condition (ii) is verifiee, and if e(Hom(R1 f∗ C, R1 g∗ C)) nest pas pu-rement of type (0, 0), then (R1 g∗ C)s ne pourrait pas be purement of type (-1, 0) + (0, -1), ce qui is absurde. On en deduit que Hom(R1 f∗ Z, R1 g∗ Z) is purement of type (0, 0), and (i) en resulte par (4.4.3) and (2.1.11.1).

Let Z a extension quadratique totalement imaginaire dun corps of nombres totalement reel Z 0 . If Z ⊗ R C ⊕ . . . ⊕ C, one defines a action par homotheties of S on Z ⊗ R en envoyant s ∈ S(R) ≃ C∗ on (s2 *
N (s)-1 , 1, . . . , 1). Le couple form of Z and of cette action of S on Z ⊗ R is a Hodge structure polarisable Zp of type (-1, 1) + (0, 0) + (1, -1), qui depend of Z and of the place choisie r : Z → C.

Let X a schema abelien on S, tel que Xe let simple. If the centre Z of End(R1 f∗ Q) nest pas totalement reel, il is extension quadratique totalement imaginaire dun corps of nombres totalement reel. If the seconde hypothese of (ii) is violee, the facteur direct R1 f∗ Q ⊗Z Zp of the famille continue of Hodge structures polarisable R1 f∗ Q ⊗Q Zp is purement of type (-1, 0) + (0, -1).

Dapres (4.4.3), the choix dun reseau entier in $R^1 f_* Q \otimes_Z Z_p$ defines a schema abelien $g : Y \to S$, and $Hom(R^1 f_* Q, R^1 g_* Q)$ nest pas purement of type (0, 0), ce qui viole (i).

Corollaire 1 (4.4.13). Let S a schema smooth connexe of point generique e and $f : X \to S$ a schema abelien on S of dimension relative $g \leq 3$. On suppose que on no revetement etale fini of S, X nadmet of sous-schema abelien constant. Then, for all schema abelien $g : Y \to S$, one a

$$Hom(X, Y) \xrightarrow{\sim} Hom(R^1 f_* Z, R^1 g_* Z)$$

and

$$Hom(Y, X) \xrightarrow{\sim} Hom(R^1 g_* Z, R^1 f_* Z).$$

Par dualite, one se ramene a ne prouver que the premiere assertion. On se ramene aussi a supposer X_e simple. Verifions que X satisfait a lune of conditions (c) of (4.4.11).

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Posons $H = M_r(D)$, D being a corps gauche of rang b^2 on son centre Z, que lon sait be totalement reel or totalement imaginaire. Posant $a = [Z : Q]$, on a

(4.4.13.1) $$ab^2 c \mid 2g \leq 6.$$

1) If Z is totalement imaginaire, and non quadratique imaginaire, one doit avoir $a = 2g = 4$. For $s \in S$, the commutant of Z agissant on $(R^1 f_* Q)_s$ is then commutatif, of sorte que laction of $\pi_1(S, s)$ is abelienne, therefore se factorise par a group fini ((4.2.8), (iii), b) or (4.2.9)). Ceci is absurde ((4.1.3.3) et (4.4.3)).

Avec the notations of (4.4.8), one pourrait aussi noter que the T_p are then of rang a, of sorte que the condition (4.4.8), b) is verifiee en chaque place and que the Hodge structure is localement constante.

2) If Z is totalement reel and if the conditions (c_1) and (c_2) are violees, then $a \geq 2$ and $b \geq 2$, ce qui is absurde.

Verifions que X satisfait aux conditions of (4.4.12). If Z is quadratique imaginaire, and sil existe $r : Z \to C$ tel que $R^1 f_* Q \otimes_{Z,r} C$ let purement of type of Hodge (-1, 0), then the Hodge structure of $R^1 f_* Q$ is localement constante, ce qui is absurde. Ceci proves (4.4.13).

(4.4.14) Let $f : X \to S$ a schema abelien polarise of dimension g on a base smooth and connexe S. If $s \in S$, the fundamental group $\pi_1(S, s)$ agit on the fibre $H_1(X_s, Z)$ of $R^1 f_* Z$ en s. La polarisation of X defines a form alternee on $H_1(X_s, Z)$, a values in Z(-1), non degeneratese on Q, and laction of $\pi_1(S, s)$ se fait en respectant cette form. Considerons lhypothese :

(4.4.14.1)

Le morphism defini par X of S in lune of composantes irreductibles of the schema of modules of varieties abeliennes polarisees correspondant is a morphism dominant.

Le result suivant ma been suggere par J.-P. Serre.

Corollaire 2 (4.4.15). Let S a schema smooth connexe and $f : X \to S$ a schema abelien polarise on S qui satisfies (4.4.14.1). Then, for all schema abelien $g : Y \to S$, one a

$$Hom(X, Y) \xrightarrow{\sim} Hom(R^1 f_* Z, R^1 g_* Z).$$

Let $s \in S$. Dapres (4.4.11) and (4.4.12), il suffices of show que the representation of $\pi_1(S, s)$ on $(R^1 f_* Q)_s$ is absolument irreductible, of sorte que H = Q. Cest ce qui resulte of the lemma suivant :

Lemme 3 (4.4.16). If X satisfies (4.4.14.1), then limage of $\pi_1(S, s)$ in the group of automorphismes symplectiques of $H_1(X_s, Z)$ is dindice fini.

Let n a entier. Quitte a remplacer S par a revetement etale surjectif, defini par a sous-groupe dindice fini of $\pi_1(S, s)$, one peut supposer que the systeme local ${}_n X = Ker(n. id : X \to X)$ is trivial on S. Remplacant X

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par a schema abelien isogene, one se ramene ensuite and of more a supposer X polarise of the serie principale. Supposons $n \geq 3$.

Le schema abelien X defines then a morphism dominant x of S in the schema of modules << of Jacobi >> M_n of schemas abeliens polarises of the serie principale munis dun isomorphism symplectique entre ${}_n X$ and $(Z/n)^{2g}$; of more, X is a image reciproque par x of the schema abelien universel on M_n. On sait que $\pi_1(M_n, x(s))$ senvoie isomorphiquement on the sous-groupe of $Sp(H_1(X_s, Z))$ kernel of lapplication of reduction mod n.

Il reste therefore a verifier que :

Lemme 4 (4.4.17). Let $p : S \to M$ a morphism dominant of schemas smooth connexes. Le sous-groupe $p_*(\pi_1(S, s))$ of $\pi_1(M, p(s))$ is then dindice fini.

Let t a point ferme of the fibre generique of p, and let T' son adherence

in S. Il existe a ouvert of Zariski dense U of M tel que T = p-1 (U) ∩ T '
let a revetement etale of U .

On ne restreint pas the generalite en prenant s ∈ T . Dans the diagramme

π1 (T, s) ----→ π1 (S, s)

ay

c

y

b

π1 (U, p(s)) ----→ π1 (M, p(s))

the fleche a a a image dindice fini, car T is a revetement fini etale, and the fleche b is surjective, car M \ U is of codimension reelle ≥ 2. La fleche c a therefore a image dindice fini.

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